

Domination number of directed tori $C_8 \times C_n^*$

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Abstract

Let $\gamma(G)$ denote the domination number of a digraph G and let C_n denote the directed cycle of length n . For $m, n \geq 2$, a 2-dimensional directed torus, denoted by $C_m \times C_n$, is the Cartesian product of C_m and C_n . Recently, the exact values of $\gamma(C_m \times C_n)$ have been derived when $2 \leq m \leq 7$ or both n and m are multiples of 3. In this paper, for $n \geq 2$, we determine the exact value

$$\gamma(C_8 \times C_n) = \begin{cases} 8 & \text{if } n = 2; \\ 16 & \text{if } n = 5; \\ 3n & \text{otherwise.} \end{cases}$$

Keyword: domination number, directed cycles, directed tori, Cartesian product,

1 Introduction

Throughout this paper, $G = (V, E)$ denotes a digraph with vertex set V and arc set E . Sometimes we use $V(G)$ instead of V and $E(G)$ instead of E if the graph G needs to be specified. Given two vertices $u, v \in V$, we say that u dominates v (or v is dominated by u) if $u = v$ or uv is an arc in E . Furthermore, for $S \subseteq V$, we say that v is dominated by S if there is a vertex $u \in S$ such that u dominates v . A subset $S \subseteq V$ is a dominating set of G if every vertex of G is dominated by S . The domination number of G , denoted by $\gamma(G)$, is the minimum cardinality of a dominating set of G . A dominating set S is minimum if $|S| = \gamma(G)$.

The Cartesian product of two digraphs $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$, denoted by $G_1 \times G_2$, is a digraph with $V_1 \times V_2$ as its vertex set and an arc $(u_1, u_2)(v_1, v_2) \in E(G_1 \times G_2)$ if and only if either $u_1v_1 \in E_1$ and $u_2 = v_2$ or $u_2v_2 \in E_2$ and $u_1 = v_1$.

*This research was partially supported by National Science Council under the Grants NSC99-2221-E-141-001, NSC98-2115-M-141-002 and NSC99-2115-M-141-001-MY2.

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Let $[n] = \{0, 1, \dots, n-1\}$ and let C_n denote the directed cycle of length n such that $V(C_n) = [n]$. For notational convenience, we simply write $T_{m,n}$ to denote the digraph $C_m \times C_n$, which is called a 2-dimensional directed torus.

It is well-known that the complexity of finding domination number on arbitrary $m \times n$ grids is still unknown. Thus, many research pursued the exact values or upper bounds of domination number for Cartesian product of paths or cycles with some restricted conditions (see [2–7] and [8]). However, to date no further research has been done for the Cartesian product of two digraphs except the recent articles described below.

Theorem 1. (Liu et al. [9].) For $n \geq 2$, $\gamma(T_{2,n}) = n$; $\gamma(T_{3,n}) = n$ if $n \equiv 0 \pmod{3}$, otherwise, $\gamma(T_{3,n}) = n + 1$; $\gamma(T_{4,n}) = \frac{3n}{2}$ if $n \equiv 0 \pmod{8}$, otherwise, $\gamma(T_{4,n}) = n + \lceil \frac{n+1}{2} \rceil$.

Theorem 2. (Zhang et al. [10].) For $n \geq 2$, $\gamma(T_{5,n}) = 2n$; $\gamma(T_{6,n}) = 2n$ if $n \equiv 0 \pmod{3}$, otherwise, $\gamma(T_{6,n}) = 2n + 2$; $\gamma(T_{m,n}) = \frac{mn}{3}$ if both n and m are multiples of 3.

Theorem 3. (Yu et al. [11].) For $n \geq 2$,

$$\gamma(T_{7,n}) = \begin{cases} \frac{5n}{2} & \text{if } n \equiv 0 \pmod{14}; \\ 2n + \lceil \frac{(n+1)}{2} \rceil & \text{if } n \equiv 3, 4, 6, 8, 10, 11 \pmod{14}; \\ 2n + \lceil \frac{(n+1)}{2} \rceil + 1 & \text{if } n \equiv 1, 2, 5, 7, 9, 12, 13 \pmod{14}. \end{cases}$$

In this paper, we determine the exact values of $\gamma(T_{8,n})$. For any terminology not given here, the reader is referred to [1].

2 Main results

For $i \in [n]$, let C_m^i denote the subdigraph of $T_{m,n}$ induced by the vertex set $\{(j, i) : 0 \leq j \leq m-1\}$, i.e., the i th column of $T_{m,n}$. From now on, unless otherwise mentioned, we assume $n \geq 8$ and the superscript in the notation C_m^i is always taken

modulo n . Moreover, all arithmetics in the first coordinate (resp. the second coordinate) of vertices are taken modulo 8 (resp. modulo n). We first give the following lower bound of $\gamma(T_{8,n})$.

Lemma 1. For $n \geq 8$, $\gamma(T_{8,n}) \geq 3n$.

Proof. Let S be a dominating set of $T_{8,n}$ and assume that $s_i = |C_8^i \cap S|$ for each $i \in [n]$. Since every vertex of C_8^i and C_8^{i+1} can respectively dominate one and two vertices in C_8^{i+1} , we have $s_i + 2s_{i+1} \geq 8$. Thus, if $s_{i+1} \leq 2$, then $s_i \geq 8 - 2s_{i+1} \geq 4$. In particular, $s_i + s_{i+1} \geq 6$ if $s_{i+1} \leq 2$. That is, either C_8^i and C_8^{i+1} have at least six vertices in S or C_8^{i+1} has at least three vertices in S . Thus, the result directly follows. $\square$

We now define the following subsets of vertices in $T_{8,n}$. For each $i \in [n]$, let

$$A_i = \{(i, i), (i + 3, i), (i + 6, i)\}.$$

Also, let $\mathbf{A}_n = \bigcup_{i \in [n]} A_i$. Obviously, $|\mathbf{A}_n| = 3n$. Note that, according to the configuration of $\mathbf{A}_n$, we have a repetition for every eight consecutive columns. For example, Figure 1 shows the case of $n = 16$. We observe that, in spite of the value of n , every vertex of $T_{8,n}$ is dominated by $\mathbf{A}_n$ except for the vertices $(2, 0)$ and $(5, 0)$. Thus, we have the following proposition.

Proposition 2. If $n \equiv 0, 3 \pmod{8}$, then $\mathbf{A}_n$ is a dominating set of $T_{8,n}$.

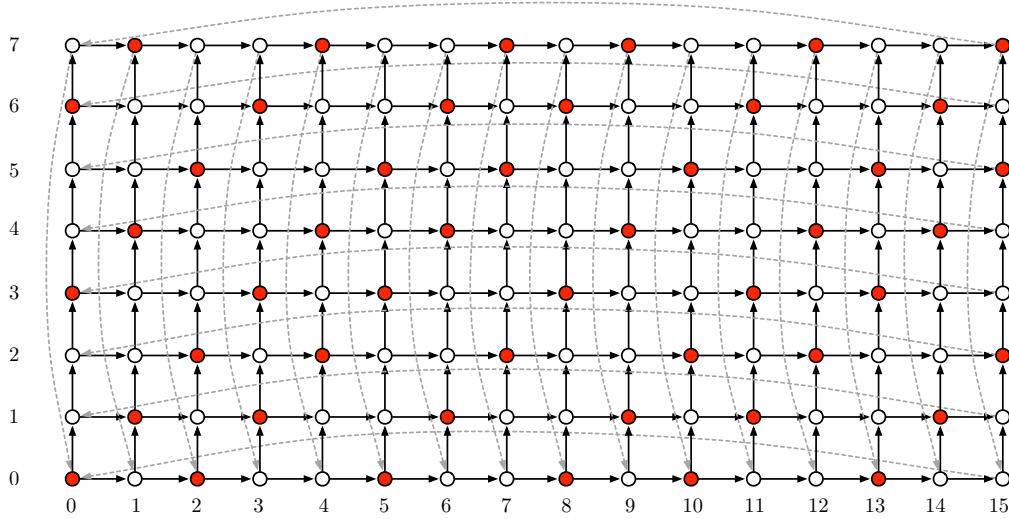


Figure 1: The set $\mathbf{A}_{16}$.

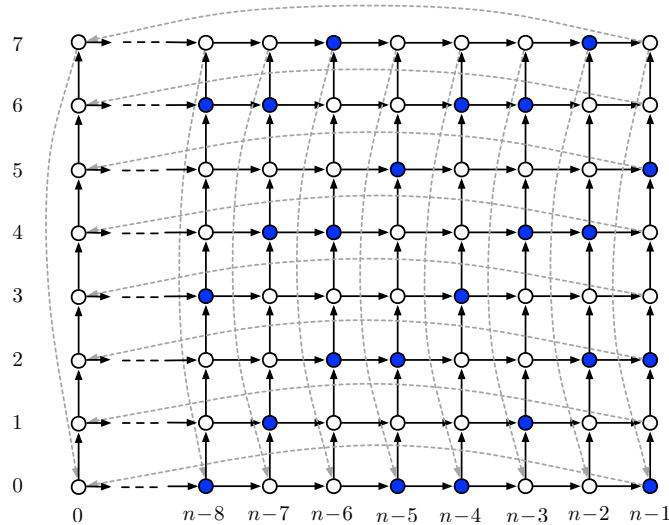


Figure 2: The set $\mathbf{B}_8$.

In order to construct a dominating set with cardinality $3n$ in $T_{8,n}$ for $n \not\equiv 0, 3 \pmod{8}$, we also define the following subsets of vertices. For each $i \in [8]$, let

$$B_i = \{(2i, n-1-i), (2i+2, n-1-i), (2i+5, n-1-i)\}.$$

Then, we let $\mathbf{B}_k = \bigcup_{i \in [k]} B_i$ where $k \in [8]$. Note that the set $\mathbf{B}_k$ designates three certain vertices in each of the last k columns of $T_{8,n}$. In particular, $(2, n-1), (5, n-1) \in \mathbf{B}_k$ for every $k \in [8]$. For example, Figure 2 shows the set $\mathbf{B}_8$. From the configuration, we can also see that every vertex in the last $k-1$ columns of $T_{8,n}$ is dominated by $\mathbf{B}_k$. The following lemmas show that combining two sets $\mathbf{A}_{n-k}$ and $\mathbf{B}_k$ for some $k \in [8]$ can obtain a dominating set of $T_{8,n}$ when $n \not\equiv 0, 3 \pmod{8}$.

Lemma 3. For $n \geq 8$, if $n \equiv 1 \pmod{8}$, then $\mathbf{A}_{n-2} \cup \mathbf{B}_2$ is a dominating set of $T_{8,n}$.

Proof. Let $S = \mathbf{A}_{n-2} \cup \mathbf{B}_2$. To check S is a dominating set of $T_{8,n}$, we need to verify that every vertex in C_8^{n-2} is dominated by S . Since $C_8^{n-3} \cap S = \{(1, n-3), (4, n-3), (6, n-3)\}$ and $C_8^{n-2} \cap S = \{(2, n-2), (4, n-2), (7, n-2)\}$ (see Figure 3 for illustration), it is clear that S is a dominating set. This completes the proof. $\square$

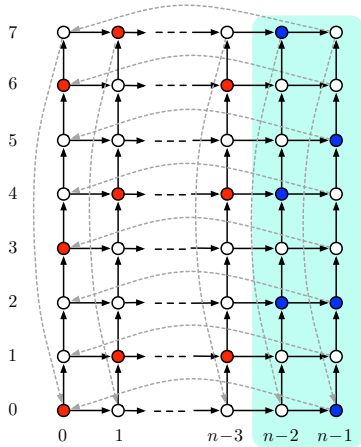


Figure 3: $\mathbf{A}_{n-2} \cup \mathbf{B}_2$ where $n \equiv 1 \pmod{8}$.

Lemma 4. For $n \geq 8$, if $n \equiv 2 \pmod{8}$, then $\mathbf{A}_{n-5} \cup \mathbf{B}_5$ is a dominating set of $T_{8,n}$.

Proof. Let $S = \mathbf{A}_{n-5} \cup \mathbf{B}_5$. To check S is a dominating set of $T_{8,n}$, we need to verify that every vertex in C_8^{n-5} is dominated by S . Since $C_8^{n-6} \cap S = \{(2, n-6), (4, n-6), (7, n-6)\}$ and $C_8^{n-5} \cap S = \{(0, n-5), (2, n-5), (5, n-5)\}$ (see Figure 4 for illustration), it is clear that S is a dominating set. This completes the proof. $\square$

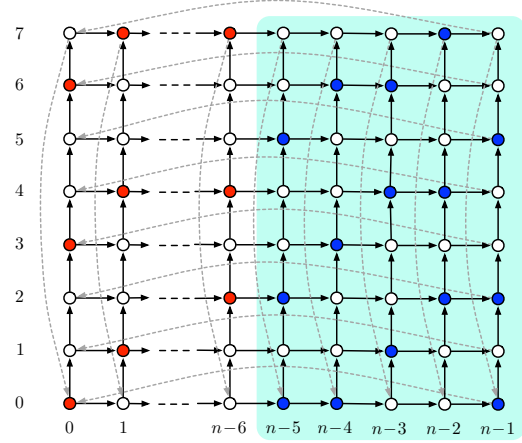


Figure 4: $\mathbf{A}_{n-5} \cup \mathbf{B}_5$ where $n \equiv 2 \pmod{8}$.

Lemma 5. For $n \geq 8$, if $n \equiv 4 \pmod{8}$, then $\mathbf{A}_{n-3} \cup \mathbf{B}_3$ is a dominating set of $T_{8,n}$.

Proof. Let $S = \mathbf{A}_{n-3} \cup \mathbf{B}_3$. To check S is a dominating set of $T_{8,n}$, we need to verify that every vertex in C_8^{n-3} is dominated by S . Since $C_8^{n-4} \cap S = \{(0, n-4), (3, n-4), (6, n-4)\}$ and $C_8^{n-3} \cap S = \{(1, n-3), (4, n-3), (6, n-3)\}$ (see Figure 5 for illustration), it is clear that S is a dominating set. This completes the proof. $\square$

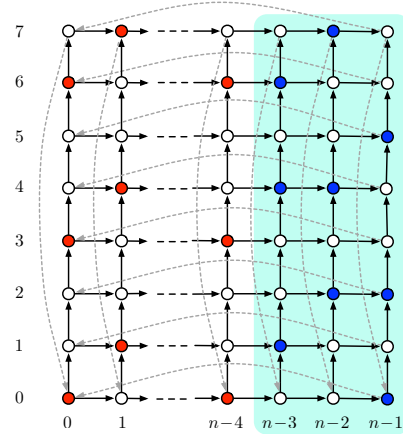


Figure 5: $\mathbf{A}_{n-3} \cup \mathbf{B}_3$ where $n \equiv 4 \pmod{8}$.

Lemma 6. For $n \geq 8$, if $n \equiv 5 \pmod{8}$, then $\mathbf{A}_{n-6} \cup \mathbf{B}_6$ is a dominating set of $T_{8,n}$.

Proof. Let $S = \mathbf{A}_{n-6} \cup \mathbf{B}_6$. To check S is a dominating set of $T_{8,n}$, we need to verify that every vertex in C_8^{n-6} is dominated by S . Since $C_8^{n-7} \cap S = \{(1, n-7), (4, n-7), (6, n-7)\}$ and $C_8^{n-6} \cap S = \{(2, n-6), (4, n-6), (7, n-6)\}$ (see Figure 6 for

illustration), it is clear that S is a dominating set. This completes the proof. $\square$

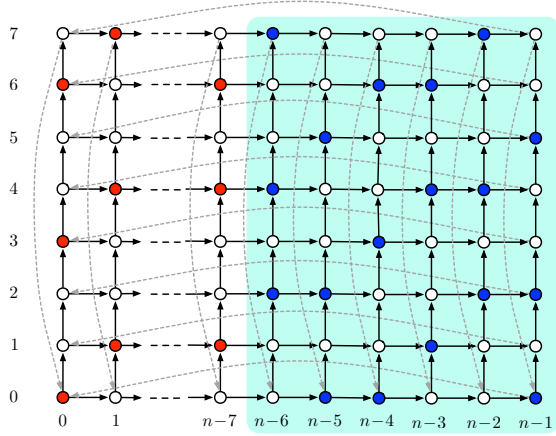


Figure 6: $\mathbf{A}_{n-6} \cup \mathbf{B}_6$ where $n \equiv 5 \pmod{8}$.

Lemma 7. For $n \geq 8$, if $n \equiv 6 \pmod{8}$, then $\mathbf{A}_{n-1} \cup \mathbf{B}_1$ is a dominating set of $T_{8,n}$.

Proof. Let $S = \mathbf{A}_{n-1} \cup \mathbf{B}_1$. To check S is a dominating set of $T_{8,n}$, we need to verify that every vertex in C_8^{n-1} is dominated by S . Since $C_8^{n-2} \cap S = \{(2, n-2), (4, n-2), (7, n-2)\}$ and $C_8^{n-1} \cap S = \{(0, n-1), (2, n-1), (5, n-1)\}$ (see Figure 7 for illustration), it is clear that S is a dominating set. This completes the proof. $\square$

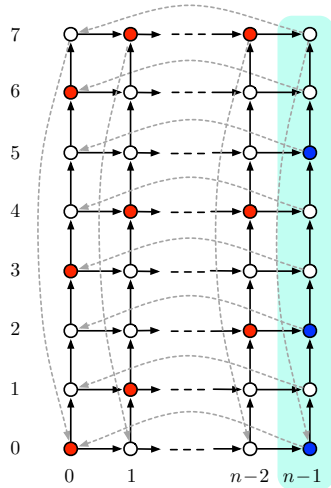


Figure 7: $\mathbf{A}_{n-1} \cup \mathbf{B}_1$ where $n \equiv 6 \pmod{8}$.

Lemma 8. For $n \geq 8$, if $n \equiv 7 \pmod{8}$, then $\mathbf{A}_{n-4} \cup \mathbf{B}_4$ is a dominating set of $T_{8,n}$.

Proof. Let $S = \mathbf{A}_{n-4} \cup \mathbf{B}_4$. To check S is a dominating set of $T_{8,n}$, we need to verify that every

vertex in C_8^{n-4} is dominated by S . Since $C_8^{n-5} \cap S = \{(0, n-5), (2, n-5), (5, n-5)\}$ and $C_8^{n-4} \cap S = \{(0, n-4), (3, n-4), (6, n-4)\}$ (see Figure 8 for illustration), it is clear that S is a dominating set. This completes the proof. $\square$

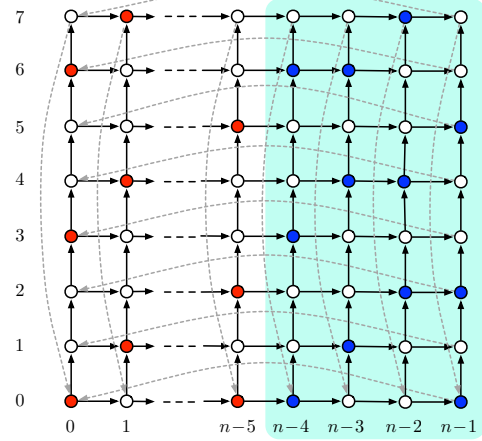


Figure 8: $\mathbf{A}_{n-4} \cup \mathbf{B}_4$ where $n \equiv 7 \pmod{8}$.

By Proposition 2 and Lemmas 3, 4, 5, 6, 7, 8, we have the following upper bound.

Lemma 9. For $n \geq 8$, $\gamma(T_{8,n}) \leq 3n$.

Therefore, according to Theorems 1, 2, 3 and Lemmas 1 and 9, we conclude the following:

Theorem 4. For $n \geq 2$,

$$\gamma(C_8 \times C_n) = \begin{cases} 8 & \text{if } n = 2; \\ 16 & \text{if } n = 5; \\ 3n & \text{otherwise.} \end{cases}$$

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